Choosing models. Explaining predictions.

AutoML, SHAP and explanation stability

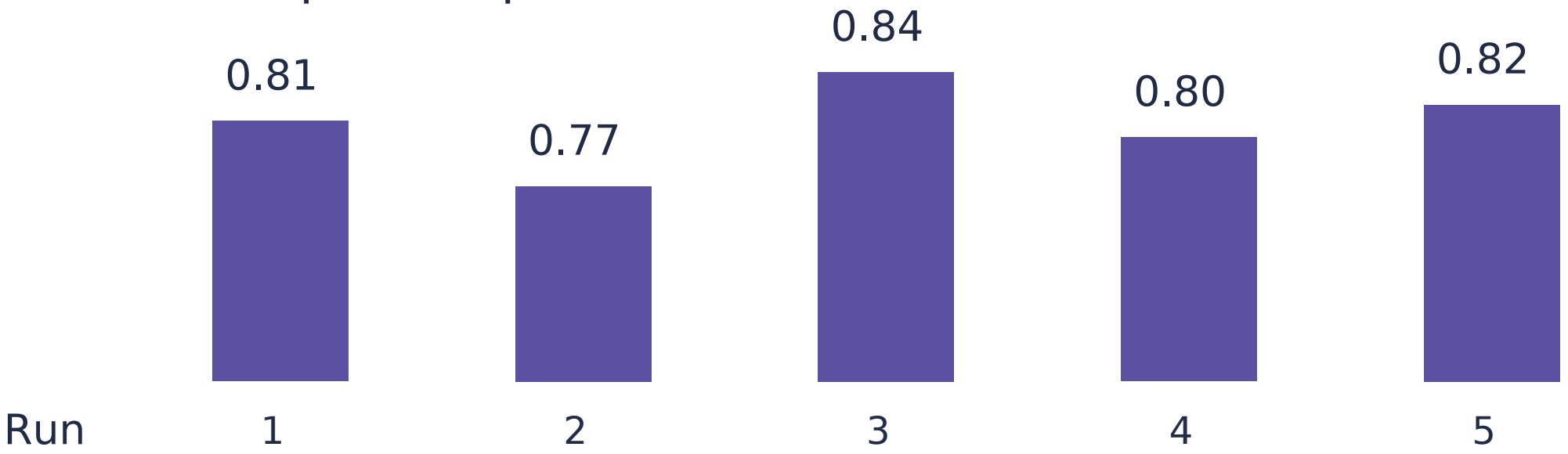
How do we choose a hypothesis function?

Why does that function make this prediction?

Last time: one run was one outcome

Illustrative repeated runs of the same experimental procedure

The seed repeats a procedure. Several runs reveal variation.



Today: what procedure is choosing the model in the first place?

First, what does one row represent?

A controlled synthetic customer example, not observed company records

One row is one customer. The target is whether that customer leaves.

The current computed example uses simulated, standardised inputs. Its values are not days, counts or months.

We must understand the inputs before interpreting a score or an explanation.

Give every input a meaning

Input	Intended meaning in the teaching example
Recency	How long since the customer last used the service
Inactivity	A duplicate of recency in this controlled simulation
Complaints	A simulated measure of customer complaints
Tenure	How long the customer has been a customer
Noise	An artificial random control, not a customer measurement

Choosing, fixing and repeating are different

Choose a configuration

Fix that configuration

Repeat training runs

Repeating training measures variation for a fixed recipe. Repeating the entire search also allows the chosen recipe to change.

State what changes: training seed, data split, search seed, or the entire procedure.

The selection rule comes before the search

Predict whether a customer will leave.

Eligible?
Mean CV recall ≥ 0.70

Among eligible candidates
Maximise mean CV precision

If no candidate meets the rule, report that. A search cannot invent an eligible solution.

We select a recipe. Training learns a function.

Feature option
Recency only

→

Tree settings
Depth 4, leaf 15

→

Fitted tree
Splits and predictions

Chosen by the search

Learned from training data

The candidate is a pipeline configuration. A fitted hypothesis function depends on training data too.

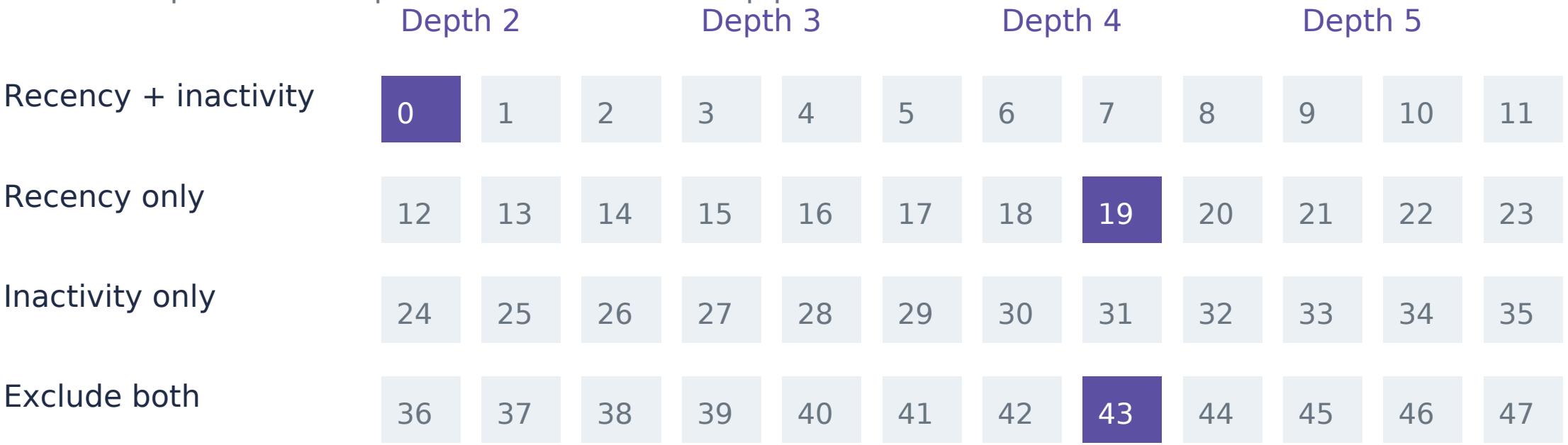
One candidate needs a fair evaluation



Fit within each fold, score on its validation portion, then average precision and recall separately.

The search space is a set of possible recipes

4 feature options × 4 depths × 3 leaf sizes = 48 pipelines



Each depth has leaf sizes 5, 15 and 30. Each highlighted square is one complete configuration.

AutoML automates parts of this selection process

We specify

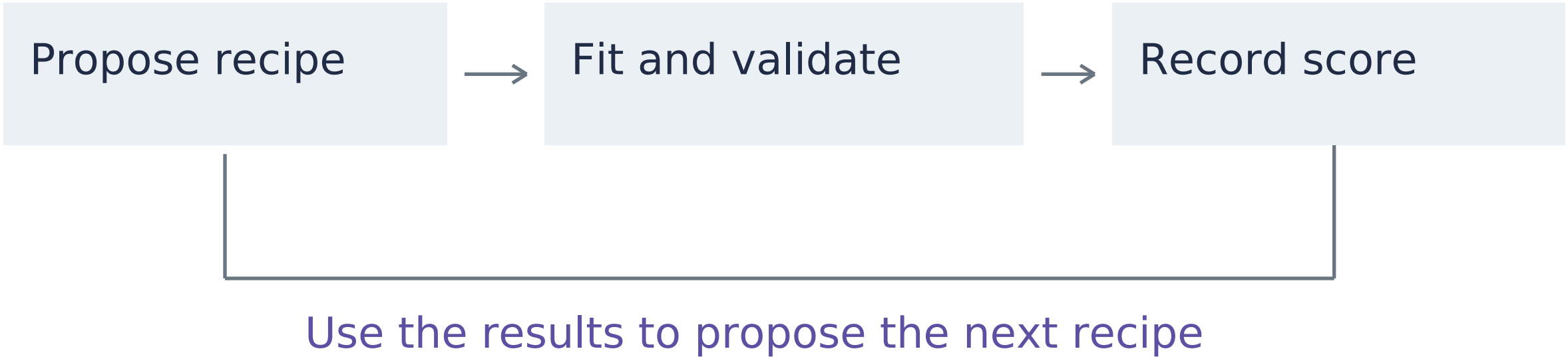
The task, available inputs, validation design, objective and budget.

The system searches

It proposes recipes, fits candidates and uses scores to decide what to try next.

AutoML can include preprocessing, feature choices, model families and hyperparameters.

The search loop surrounds model fitting



Stop at the declared budget. Keep the best eligible candidate found.

Frameworks package the machinery

Examples of different design choices

Framework	Search idea	What it helps organise
auto-sklearn	Bayesian optimisation	Pipeline search and ensembles
TPOT[1]	Evolutionary search	Pipeline construction and evaluation

[1] TPOT: Tree-based Pipeline Optimization Tool. Defaults still express modelling choices.

Automation inherits our decisions and mistakes

If we supply...	The consequence can be...
A poor metric	A strong score for the wrong goal
An incomplete search space	No access to a useful candidate
Leaked or unsuitable validation	Efficient selection of misleading results
A small budget	A result shaped by the visited candidates

Simulated annealing explores one candidate at a time

Current recipe A



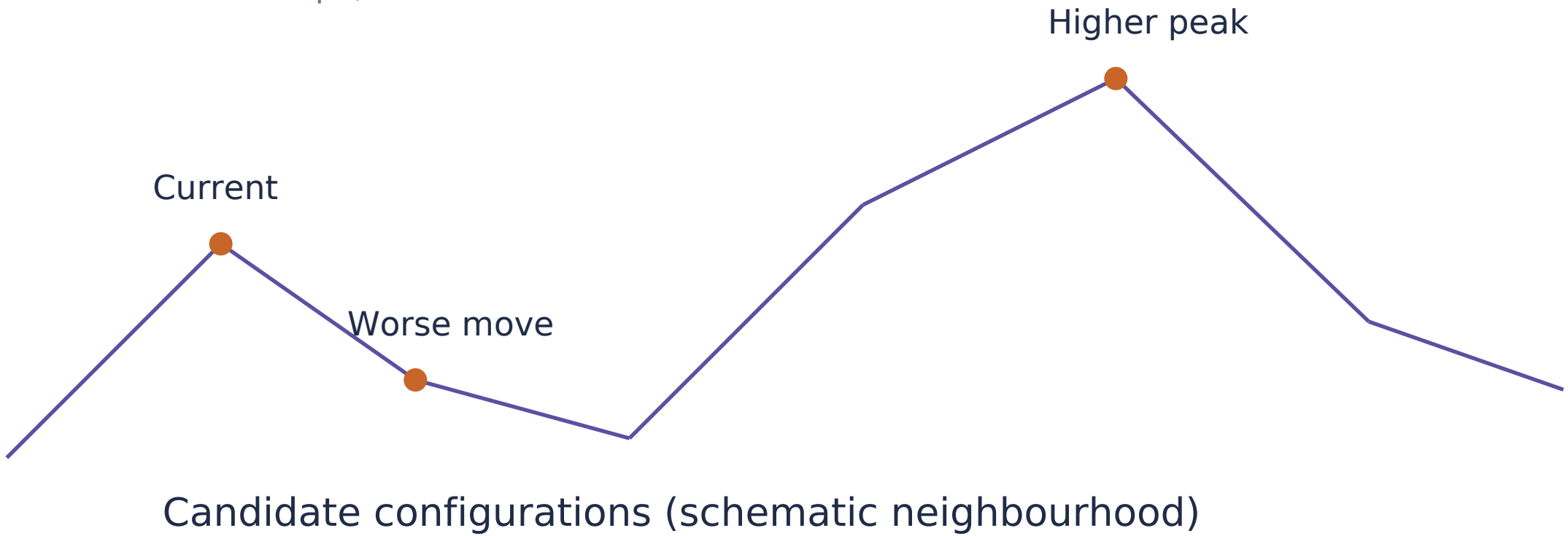
Proposed neighbour B

Change one choice, such as tree depth 3 to 4. Train and validate the new recipe using the same folds.

Keep the best eligible candidate seen, even when the current candidate moves elsewhere.

Sometimes a worse move helps escape a local peak

Schematic score landscape, not measured model results



A greedy search rejects every worse move. Annealing can cross a lower-scoring region.

Cooling makes worse moves less likely

For two eligible candidates, let the score be CV precision.

Precision decreases by 0.02	Temperature T	Accept with probability
Early exploration	0.05	exp(-0.02 / 0.05) ≈ 67%
Later exploration	0.005	exp(-0.02 / 0.005) ≈ 2%

Better moves are accepted. Define feasibility handling, cooling schedule and budget in advance.

Why consider evolutionary search?

The choices are mixed

Feature subsets, model families and integer settings do not form one smooth surface.

A population explores alternatives

Selection uses validation feedback. Variation creates new recipes.

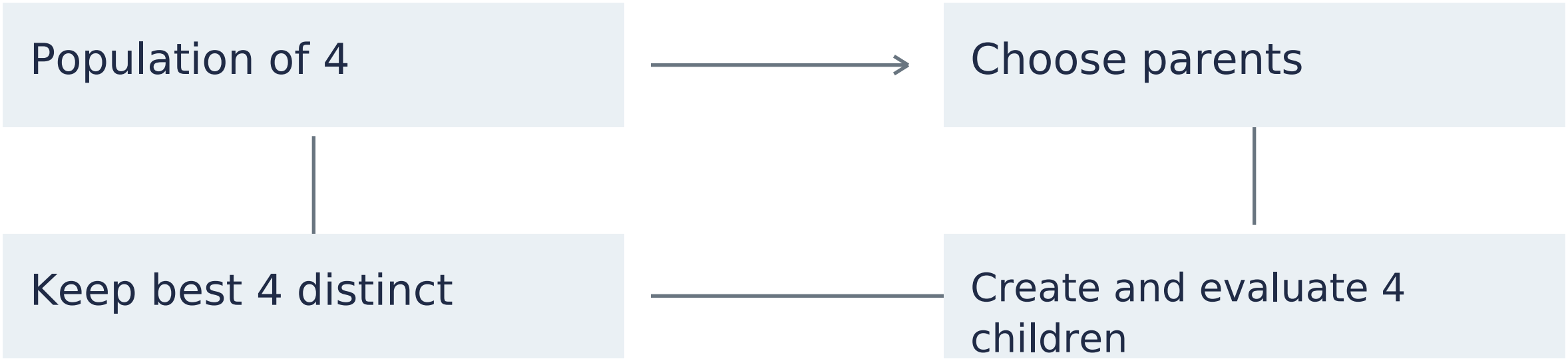
Evolution still costs evaluations, can converge early and needs comparison with simple baselines.

A candidate can be encoded as a chromosome

Feature choice	Maximum depth	Minimum leaf size
Recency only	4	15

Changing a gene changes the recipe. We must train and evaluate the new candidate.

Selection, variation, evaluation, survival



A population is a set of candidate pipelines. A generation is one round of this loop.

Recombination and mutation change the recipe

Candidate	Features	Depth	Leaf
Parent A	Recency only	4	15
Parent B	Recency + inactivity	2	30
Recombined child	Recency only	2	30
After mutation	Recency only	5	30

A search result depends on its budget and seed

Hold fixed

Candidate space, folds, scoring rule and evaluation budget.

Repeat the search

Change search seeds. Record outcomes, failures and evaluation costs.

This 48-candidate demo explains the mechanism. It cannot establish that evolution is superior.

A good score leaves another question

Why did this fitted model make this prediction?

Customer 1

Selected model → Leave: 90.8%

An explanation describes how the model uses information to produce an output.

Different questions need different explanations

Question	Scope
Why this prediction for customer 1?	Local explanation
Which inputs receive large attributions across customers?	Global summary
Would changing customer support prevent leaving?	Causal question

A prediction needs a reference for comparison

Reference prediction
46.8%

Average output on 60 background customers

Customer 1 prediction
90.8%

Output for this particular input

Which input features receive credit for the difference?

SHAP allocates a prediction difference

SHapley Additive exPlanations

Reference output + feature contributions = model output

Each contribution averages what a feature adds when it joins different groups of other features.

What does an absent feature mean?

Keep customer values
for present features

Use background values
for absent features

Predict for each completed input, then average the outputs.

The fitted model stays fixed. The background and replacement rule define the explanation question.

Two features let us calculate the idea by hand

Illustrative coalition values, independent of the larger model

Features revealed	Average predicted probability
Neither	30%
A only	50%
B only	40%
A and B	70%

Reveal A first, then B

Neither
30%

→

A only
50%

→

A and B
70%

A adds 20 points

B adds 20 points

Reveal B first, then A



B adds 10 points

A adds 30 points

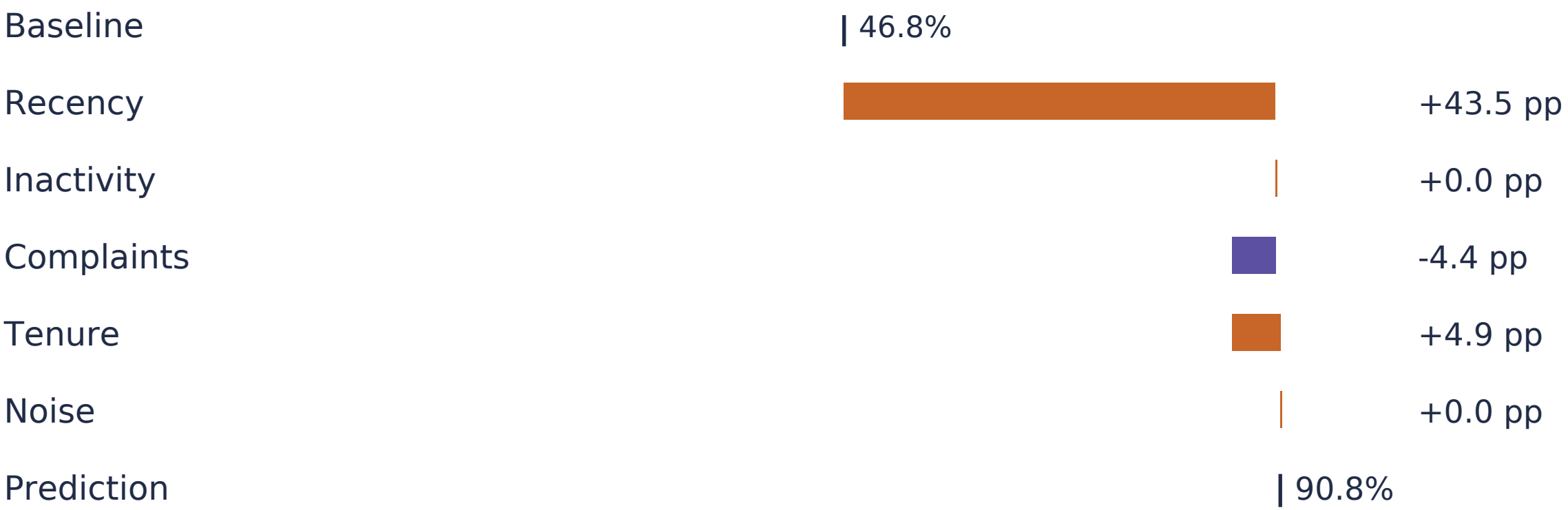
Shapley values average across the orders

Feature	Marginal contributions	Average
A	20 and 30 points	+25 points
B	20 and 10 points	+15 points

30% + 25 points + 15 points = 70%

A waterfall connects the baseline to the prediction

Actual exact Shapley values for M19 and customer 1



A negative contribution lowers this model output

Positive SHAP value

Pushes this prediction above the reference output.

Negative SHAP value

Pushes this prediction below the reference output.

A contribution sign does not tell us the effect of intervening on a real customer.

Local explanations can be summarised globally

Same model, same background, 100 audit customers



Mean absolute SHAP measures typical attribution magnitude. Taking absolute values discards direction.

A SHAP value needs its context

Record	In this example
Fitted model and output	M19, probability of leaving
Case or sample	Customer 1, or 100 audit customers
Background	60 fixed development customers
Absent-feature rule	Empirical interventional replacement

A precise explanation still has limits

Inputs can overlap

Replacing one correlated input may create unusual combinations.

The question can change

A different background or dependence assumption can change the attribution.

SHAP explains a model output under stated assumptions. It does not prove causality or model validity.

An AI partner can help us check a claim

“AutoML found the best model. SHAP proves that inactivity causes customers to leave.”

What evidence is missing from each sentence?

Ask the AI partner for a counterexample, then check it against the search and explanation ledger.

What we can now explain

The search selects a pipeline configuration using validation evidence.

SHAP allocates a fixed model output relative to a declared reference.

Now a research question: would another similarly performing model tell the same story?

Research glimpse: one good model may not be unique

	Depth 2		Depth 3		Depth 4		Depth 5					
Recency + inactivity	0	1	2	3	4	5	**6**	**7**	8	9	**10**	11
Recency only	12	13	14	15	16	17	**18**	**19**	20	21	**22**	23
Inactivity only	24	25	26	27	28	29	**30**	**31**	32	33	**34**	35
Exclude both	36	37	38	39	40	41	42	43	44	45	46	47

Several candidate pipelines can fit nearly as well under the same evaluation rule.

The Rashomon idea keeps near-optimal alternatives

Keep eligible candidates within 2 precision points of the best tested eligible candidate.

Best observed	Cutoff	In the set
83.29%	81.29%	9 of 48

Similar scores can support different attributions

M19: Recency only

Precision 83.25%
Prediction 90.8%

Recency: +43.5 points

Inactivity: +0.0 points

M31: Inactivity only

Precision 83.25%
Prediction 90.8%

Recency: +0.0 points

Inactivity: +43.5 points

Same customer, background and SHAP definition. Only the fitted model changes.

Stability requires saying what varies

Change	Keep fixed to isolate it
Training sample or seed	Recipe, audit cases, explanation settings
Selected model	Audit cases, background, SHAP definition
Background sample	Fitted model and cases

Persistent attribution is evidence of robustness to that change. It does not establish causal truth.

The whole experiment needs an honest claim

We searched a declared space under a fixed budget.

We explained a specified output with a specified reference.

We checked whether the explanation persisted across alternatives.

Your research connection: dependable predictions and dependable explanations are related questions.

Sources for AutoML and explanation

Olson and Moore (2016). TPOT: A Tree-based Pipeline Optimization Tool.

https://proceedings.mlr.press/v64/olson_tpot_2016.html

Lundberg and Lee (2017). A Unified Approach to Interpreting Model Predictions.

https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Fisher, Rudin and Dominici (2019). All Models are Wrong, but Many are Useful.

https://jmlr.org/papers/v20/18-760.html

Example methods and further reading

Synthetic independent customers. Shared 3-fold CV on 900 development cases. Fixed tree seed. A separate audit set supports exploratory explanations. The reserved test set stays unused.

Exact Shapley values enumerate 32 feature subsets using 60 fixed background rows. Local additivity is checked for every model and audit case.

Framework documentation: automl.github.io/auto-sklearn/master/manual.html
TPOT: epistasislab.github.io/tpot/latest/
SHAP: shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html